

Demonstration of Tests of Conversion of PM Dot Notation to Parentheses

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SECTION 0. VERIFICATION TESTS (of dot to paren dot icn)

For each proposition is given:

- 1: the PM notation with dots.
 - 2: the notation with parentheses
 - 3: the Polish (with Lukasiewicz symbols) notation
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*2•06⊢ $p \supset q \supset q \supset r \supset p \supset r$

Version with parentheses

*2•06⊢ $(p \supset q) \supset ((q \supset r) \supset (p \supset r))$

Polish Lukasiewicz notation

*2•06⊢ $CCpqCCqrCpr$

*3•47⊢ $p \supset r \supset q \supset s \supset p \supset q \supset r \supset s$

Version with parentheses

*3•47⊢ $(p \supset r) \wedge (q \supset s) \supset ((p) \wedge (q) \supset (r) \wedge (s))$

Polish Lukasiewicz notation

*3•47⊢ $CKCprCqsKCKpqrs$

*4•22⊢ $p \equiv q \supset q \equiv r \supset p \equiv r$

Version with parentheses

*4•22⊢ $(p \equiv q) \wedge (q \equiv r) \supset (p \equiv r)$

Polish Lukasiewicz notation

*4•22⊢ $CKEpeqEqrEpr$

*4•41⊢ $p \vee q \supset r \equiv p \vee q \supset p \vee r$

Version with parentheses

*4•41⊢ $((p) \vee (q) \wedge (r)) \equiv (p \vee q) \wedge (p \vee r)$

Polish Lukasiewicz notation

*4·41 $\vdash$ $KEKApqrApqApr$

*4·43 $\vdash$ $\vdash p \equiv \vdash p \vee q \cdot p \vee \sim q$

Version with parentheses

*4·43 $\vdash$ $(p) \equiv ((p \vee q) \wedge (p \vee \sim q))$

Polish Lukasiewicz notation

*4·43 $\vdash$ $EpKApqApNq$

*4·44 $\vdash$ $\vdash p \equiv \vdash p \cdot \vee \cdot p \cdot q$

Version with parentheses

*4·44 $\vdash$ $(p) \equiv ((p) \vee (p) \wedge (q))$

Polish Lukasiewicz notation

*4·44 $\vdash$ $EpKAppq$

*4·87 $\vdash$ $\vdash p \cdot q \cdot \supset \cdot r \equiv \vdash p \cdot \supset \cdot q \supset r \equiv \vdash q \cdot \supset \cdot p \supset r \equiv \vdash q \cdot p \cdot \supset \cdot r$

Version with parentheses

*4·87 $\vdash$ $((p) \wedge (q) \supset (r)) \equiv ((p) \supset (q \supset r)) \equiv ((q) \supset (p \supset r)) \equiv ((q) \wedge (p) \supset (r))$

Polish Lukasiewicz notation

*4·87 $\vdash$ $EEECKpqrCpCqrCqCprCKqpr$

*4·88 $\vdash$ $\vdash p \cdot q \cdot \supset \cdot r \equiv \vdash p \cdot \supset \cdot q \supset r \equiv \vdash q \cdot \supset \cdot p \supset r \equiv \vdash q \cdot p \cdot \supset \cdot r$

Version with parentheses

*4·88 $\vdash$ $(p) \wedge (q) \supset (r) \equiv ((p) \supset (q \supset r)) \equiv ((q) \supset (p \supset r)) \equiv ((q) \wedge (p) \supset (r))$

Polish Lukasiewicz notation

*4·88 $\vdash$ $EEECKpqrCpCqrCqCprCKqpr$

*5·33 $\vdash$ $\vdash p \cdot q \supset r \equiv \vdash p \cdot p \cdot q \cdot \supset \cdot r$

Version with parentheses

*5·33 $\vdash$ $(p) \wedge (q \supset r) \equiv (p) \wedge ((p) \wedge (q) \supset (r))$

Polish Lukasiewicz notation

*5·33 $\vdash$ $KEKpCqrpCKpqr$

From Landon D. C. Elkind's Paper in Russell: Vol. 43, no. 1, page 44

*431·441 $\vdash$ $\vdash p \vee q \equiv \vdash r \supset s$

Version with parentheses

*431·441 $\vdash$ $(p \vee q) \equiv (r \supset s)$

Polish Lukasiewicz notation

*431·441 $\vdash$ $EApqCrs$

*431·442 $\vdash$ $\vdash p \cdot \vee \cdot q \equiv \vdash r \cdot \supset \cdot s$

Version with parentheses

*431·442 $\vdash$ $((p) \vee (q \equiv r)) \supset ((s))$

Polish Lukasiewicz notation

*431·442 $\vdash$ $CpEqrs$

*431·443 $\vdash p \vee q \equiv r \supset s$

Version with parentheses

*431·443 $\vdash ((p \vee q) \equiv (r)) \supset ((s))$

Polish Lukasiewicz notation

*431·443 $\vdash CEApqrs$

*431·444 $\vdash p \vee q \equiv r \supset s$

Version with parentheses

*431·444 $\vdash (p) \vee ((q \equiv r) \supset (s))$

Polish Lukasiewicz notation

*431·444 $\vdash ApCEqrs$

*431·445 $\vdash p \vee q \equiv r \supset s$

Version with parentheses

*431·445 $\vdash (p) \vee ((q \equiv (r \supset s)))$

Polish Lukasiewicz notation

*431·445 $\vdash ApEqCr s$

From same, page 54

*431·54 $\vdash p \cdot q \cdot r \cdot s \supset p \cdot s \cdot r \cdot q$

Version with parentheses

*431·54 $\vdash ((p) \wedge (q)) \wedge ((r) \wedge (s)) \supset ((p) \wedge (s)) \wedge ((r) \wedge (q))$

Polish Lukasiewicz notation

*431·54 $\vdash KCKKpqKrsKpsKrq$

check longer prop name

Version with parentheses

Polish Lukasiewicz notation

Propositions involving quantifiers

*9·2 $\vdash (x) \cdot psix \cdot \supset \cdot psiy$

Version with parentheses

*9·2 $\vdash (((x))psix) \supset (psiy)$

Polish Lukasiewicz notation

*9·2 $\vdash C(x)psixpsiy$

*9·21 $\vdash \cdot (x) \cdot psix \supset phix \cdot \supset \cdot (x) \cdot psix \cdot \supset \cdot (x) \cdot phix$

Version with parentheses

*9·21 $\vdash (((x))psix \supset phix) \supset (((x))psix) \supset (((x))phix)$

Polish Lukasiewicz notation

*9·21 $\vdash CCC(x)psixphix(x)psix(x)phix$

*9·22 $\vdash \cdot (x) \cdot psix \supset phix \cdot \supset \cdot (\mathfrak{A}x) \cdot psix \cdot \supset \cdot (\mathfrak{A}x) \cdot phix$

Version with parentheses

*9·22 $\vdash (((x))psix \supset phix) \supset (((\mathfrak{A}x))psix) \supset (((\mathfrak{A}x))phix)$

Polish Lukasiewicz notation

*9·22 $\vdash CCC(x)psixphix(\mathfrak{A}x)psix(\mathfrak{A}x)phix$

*9·31 $\vdash ::(\mathfrak{A}x).phix.\vee.(\mathfrak{A}x).phix:\supset.(\mathfrak{A}x).phix$

Version with parentheses

*9·31 $\vdash (((((\mathfrak{A}x))phix) \vee ((\mathfrak{A}x))phix)) \supset ((\mathfrak{A}x))phix$

Polish Lukasiewicz notation

*9·31 $\vdash CA(\mathfrak{A}x)phix(\mathfrak{A}x)phix(\mathfrak{A}x)phix$

*9·401 $\vdash ::p:\vee:q.\vee.(\mathfrak{A}x).psix::\supset::q:\vee:p.\vee.(\mathfrak{A}x).psix$

Version with parentheses

*9·401 $\vdash (((p) \vee ((q) \vee ((\mathfrak{A}x))psix))) \supset ((q) \vee ((p) \vee ((\mathfrak{A}x))psix))$

Polish Lukasiewicz notation

*9·401 $\vdash CApAq(\mathfrak{A}x)psixAqAp(\mathfrak{A}x)psix$

*10·35 $\vdash ::(\mathfrak{A}x).p.psix.\equiv:p:(\mathfrak{A}x).psix$

Version with parentheses

*10·35 $\vdash (((\mathfrak{A}x))p) \wedge (psix) \equiv (p) \wedge (((\mathfrak{A}x))psix)$

Polish Lukasiewicz notation

*10·35 $\vdash KEK(\mathfrak{A}x)ppsixp(\mathfrak{A}x)psix$

*11·2 $\vdash :(x, y).phi[x, y].\equiv.(y, x).phi[x, y]$

Version with parentheses

*11·2 $\vdash ((x, y).phi[x, y]) \equiv ((y, x).phi[x, y])$

Polish Lukasiewicz notation

*11·2 $\vdash E(x, y).phi[x, y](y, x).phi[x, y]$

One Step in proof of 11.55 I wanted example of 2 adjacent quantifiers -
hard to find.

*11·551 $\vdash ::(x)::(\mathfrak{A}y).psix.phi[x, y].\equiv:psix:(\mathfrak{A}y).phi[x, y]$

Version with parentheses

*11·551 $\vdash (((x))(\mathfrak{A}y))psix) \wedge (phi[x, y]) \equiv (psix) \wedge (((\mathfrak{A}y))phi[x, y])$

Polish Lukasiewicz notation

*11·551 $\vdash KEK(x)(\mathfrak{A}y)psixphi[x, y]psix(\mathfrak{A}y)phi[x, y]$

From same, page 46

*431·46 $\vdash :(x).psix.phix.\supset.(x).psix$

Version with parentheses

*431·46 $\vdash (((x))psix) \wedge (phix) \supset ((x))psix$

Polish Lukasiewicz notation

*431·46 $\vdash CK(x)psixphix(x)psix$

Other Tests

*99·99 $\vdash ::\sim(\mathfrak{A}x):\sim psix.\supset.(x).\sim psix$

Version with parentheses

*99•99⊢ $((\sim(\mathfrak{A}x))\sim_{psix}) \supset ((x))\sim_{psix}$

Polish Łukasiewicz notation

*99•99⊢ $CN(\mathfrak{A}x)N_{psix}(x)N_{psix}$